

LECTURE 3: EVALUATION

METRICS THAT MATTER FOR MEDICAL IMAGING

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August 30, 2026

Evaluation, and what to expect

TERMINOLOGY

		Prediction	
		0	1
Ground truth	0	TN m_{00}	FP m_{01}
	1	FN m_{10}	TP m_{11}

↑ missed disease

Be careful: some packages transpose this layout.

- ▶ **True positive** (m_{11}): an example of class 1 predicted as class 1.
- ▶ **False positive** (m_{01}): class 0 predicted as class 1. **Type I error.**
- ▶ **True negative** (m_{00}): class 0 predicted as class 0.
- ▶ **False negative** (m_{10}): class 1 predicted as class 0. **Type II error.**

Total instances $m = m_{00} + m_{01} + m_{10} + m_{11}$, and the **error rate** is $(m_{01} + m_{10})/m$.

Error rate hides imbalance. If 10% of examples are class 1 and 90% are class 0, calling *everything* class 0 already scores a 10% error rate. On a disease with 5% prevalence, always predicting “healthy” reaches **95% accuracy** and catches **zero** patients: every error it makes is an m_{10} , the one cell that matters clinically.

COMMON MEASURES

What machine learning calls them

- ▶ Accuracy = $\frac{TP + TN}{TP + FP + FN + TN}$
- ▶ Precision = $\frac{TP}{TP + FP}$ of the declared positives
- ▶ Recall = $\frac{TP}{TP + FN}$ of the actual positives
- ▶ $F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

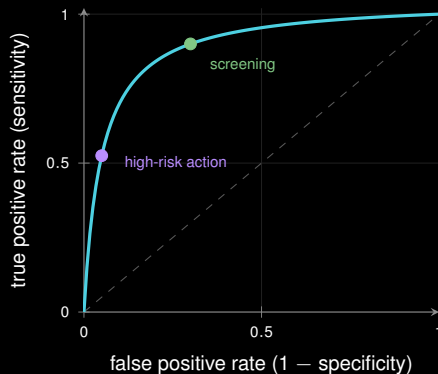
What medicine calls them

- ▶ Sensitivity = $\frac{TP}{TP + FN}$ the same as recall
- ▶ Specificity = $\frac{TN}{FP + TN}$ of the actual negatives
- ▶ False positive rate = $\frac{FP}{FP + TN}$
- ▶ PPV = precision, NPV = $\frac{TN}{TN + FN}$

Two literatures, one set of numbers

Sensitivity is recall. Text classification reports *precision* and *recall*; medicine reports *PPV* and *sensitivity*. And none of these is a fixed property of the model: each one moves with the **threshold** you pick for calling something positive.

ROC, AUC, AND CHOOSING WHERE TO SIT ON THE CURVE



The curve

Sweep the threshold from 1 to 0 and plot sensitivity against 1—specificity. **AUC** summarizes the whole curve in one number: threshold-free, good for comparing models.

The operating point

AUC does not deploy. You must pick a point: **high sensitivity** for screening (a miss is worse than a callback), **high specificity** before a biopsy or surgery.

Two footnotes

Youden's index ($J = \text{sens} + \text{spec} - 1$) picks a balanced point when you have no preference. At very low prevalence, prefer the **precision–recall** curve.

PITFALLS, AND WHAT TO EXPECT

Overfitting

Train accuracy $\gg$ validation accuracy. Reach for augmentation, weight decay, early stopping, and pretrained weights, in that order.

Class imbalance

The rare class gets ignored and the loss looks fine. Use class weights or resampling, and always read *per-class* metrics.

Leakage

The same patient in train and test inflates every number you report. Split by patient. Check it explicitly: do not assume the dataset did it.

What to expect

On a well-posed task with a few thousand images per class, a fine-tuned ResNet-18 reaches accuracy in the high 80s to low 90s (%) within a few epochs, a clear margin over the same network trained from scratch.